

Synthetic data-driven explainable machine learning for groundwater salinity prediction in the Al-Qatif coastal aquifer of Saudi Arabia

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ABSTRACT

Study region: The Al-Qatif coastal aquifer, Saudi Arabia.

Study focus: Groundwater salinity is a concern in coastal aquifers, where limited data restrict predictive modeling. Machine learning (ML) models have shown promise for salinity assessment; however, their performance is constrained by small sample sizes. To address this limitation, groundwater salinity was modeled using an integrated ML framework. A dataset of thirty-nine groundwater samples was augmented with 700 synthetic samples. Salinity (mg/L) was used as the target variable, and the synthetic data quality was evaluated using Jensen–Shannon Divergence (JSD), Maximum Mean Discrepancy (MMD), and Charge Balance Error (CBE). Four ML models were trained on both datasets.

New hydrologic insights: The results showed that Gaussian Mixture Models (GMMs) can preserve both the statistical structure and hydrochemical behavior of groundwater salinity data under data scarcity. The generated synthetic samples exhibited low marginal divergence from observed data with average JSD of 0.0797 and no detectable difference in joint multivariate structure as shown by MMD value of 0.0. The CBE confirmed that ionic balance characteristics were maintained rather than artificially enforced. Among the tested models, the Gradient Boosting Machine (GBM) demonstrated the most consistent generalization to real-only test data (Relative Root Mean Squared Error: rRMSE = 7.87 % and Relative Mean Absolute Error: rMAE = 6.73 %). Explainable analysis identified bromide, sodium, and chloride as the dominant factors controlling groundwater salinity in the study area.

1. Introduction

Groundwater is a treasure of natural resources and supports over 2 billion people worldwide in drinking water supply, agriculture, and industry (Ahmadi et al., 2022). It is essential for the sustainability of surface water systems through stream and wetland recharge

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(Li et al., 2019). However, coastal aquifer groundwater is especially at increased risk of salinization from increased anthropogenic processes (Jibrin et al., 2025; Schiavo et al., 2023; Slama et al., 2020). Seawater intrusion from hydraulic head imbalances draws down freshwater reserves (Nyika et al., 2024), mainly from unconfined aquifers lacking protective barriers. Over-abstraction further worsens salinity problems and lowers water quality and suitability (Aldrees et al., 2025b; Ondrasek and Rengel, 2021). Salinization threatens farming productivity and reduces economic viability (Colombani et al., 2024). These issues are specifically severe in regions where dependence on groundwater amplifies vulnerability (Agbasi et al., 2025; Bouselsal et al., 2024; Pande et al., 2025). Sea-level rise, intensive abstraction, and climate-driven hydrologic changes collectively emphasize the need for improved management.

Groundwater salinity is the accumulation of dissolved salts and is a major indicator of water quality deterioration because of sewer intrusion in coastal aquifers. It occurs when over-extraction lowers groundwater levels and allows saline water to advance inland (Cantelon et al., 2022; Hussain et al., 2019). Traditional assessment techniques such as hydrochemical analysis (Nair et al., 2021), geophysical techniques (Hoffmann et al., 2025), physically based models (Taherian et al., 2021), and data-driven prediction models (Sarkar et al., 2024), offer valuable insights but face practical constraints. Hydrochemical approaches reveal governing processes (Qiu et al., 2024) but could lack temporal resolution (Kazakis et al., 2016). Physically based models can capture aquifer dynamics yet require intensive calibration and are numerically demanding (Di Salvo, 2022). Geophysical methods, which offer non-invasive imaging, were known to be limited by subsurface heterogeneity, limiting their sensitivity (El-Sayed et al., 2023). These limitations require different approaches that combine predictive power with scalability.

Recent advances in machine learning (ML) offer a promising path for evaluating complex and nonlinear relationships in environmental research (Jibrin et al., 2024; Usman et al., 2025a). ML techniques have been applied to groundwater levels (Zhu et al., 2025), groundwater heavy metal pollution (Li et al., 2024), mapping groundwater potential zones (Rana et al., 2025), groundwater vulnerability assessment (Raisa et al., 2024), and predicting groundwater quality (Jibrin et al., 2024). For salinity studies, ML has demonstrated flexibility across different hydrogeological contexts. Examples include Ridge Regression and ensemble models for seawater intrusion in Saudi aquifers (BinMakhshen and Benaafi, 2024), Extreme Gradient Boosting (XGBoost) and Random Forest (RF) in China's terminal lakes (Xu et al., 2024), Neural Networks in Tunisia (Kraiem et al., 2024), hybrid deep-learning optimizers in Vietnam (Nguyen et al., 2025), and ensemble frameworks in the USA (Mahmoud et al., 2025). Other applications include Adaptive Neuro-Fuzzy Inference Systems (ANFIS) in Saudi Arabia (Abba et al., 2022) and Categorical Boosting (CatBoost) regression in Vietnam's Mekong Delta (Tran et al., 2021). Together, these studies highlight that ML can handle complex and nonlinear patterns. However, their performance depends on the choice of algorithm and data availability. The Al-Qatif region of eastern Saudi Arabia provides a representative case of these problems. Groundwater supports agricultural, industrial, and residential uses, yet swift urbanization, land reclamation, and industrial growth have increased extraction pressures.

Limited availability of groundwater observations is a major constraint on the reliable application of ML models. Several groundwater studies that apply ML for salinity prediction have largely relied on limited measured datasets to train and validate models without augmentation (Boudibi et al., 2024; Jeong et al., 2024). Where synthetic data have been introduced, deep generative models are adopted despite their requirements of large training data to achieve stable learning. On the other hand, Gaussian Mixture Models (GMMs) offer a statistically grounded and data-efficient alternative that explicitly models multivariate dependence (Gomes et al., 2022; Maurer et al., 2021). The use of GMM-based synthetic data generation for groundwater salinity prediction, combined with detailed validation as well as interpretability analysis, remains unexplored.

This study tackles the limitations of data-scarce groundwater salinity modeling through a combined framework of synthetic data generation, ML, and model explainability. Small sample sizes are known to undermine the stability together with generalization of ML predictions in hydrogeological applications (Xue et al., 2025). To address the limitations, GMMs have been proposed to produce synthetic groundwater salinity information which retains statistical properties and multivariate dependence properties characteristic of these measurements. In contrast to deep generative techniques, Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), which depend on large training sets that lack transparency in small sample sizes, GMMs offer an empirical, data-efficient method that is not limited by small sample sizes and is especially capable of handling even imbalanced groundwater data. The generated synthetic data is validated using Jensen–Shannon Divergence (JSD) to evaluate the consistency of the distribution before model training. Maximum Mean Discrepancy (MMD) is used to assess the multivariate similarity between the actual and synthetic datasets and to evaluate the preservation of joint dependence among hydrochemical variables. Moreover, Charge Balance Error (CBE) analysis is performed to assess hydrochemical consistency, ensuring that the synthetic samples maintain ionic balance characteristics similar to those of the real groundwater measurements. Following the validation, four ML models, Light Gradient Boosting Machine (LightGBM), XGBoost, RF, and Gradient Boosting Machine (GBM), are applied for groundwater salinity prediction, and SHapley Additive exPlanations (SHAP) are used to examine the contribution of individual hydrochemical variables. The proposed framework provides a transparent, methodologically controlled approach to groundwater salinity modeling in data-limited settings.

2. Study area and experimental data collection

2.1. Geological and hydrogeological setting of Al-Qatif

The Al-Qatif coastal zone lies along a low-lying plain bordering the Arabian Gulf and includes Al-Qatif City and Tarout Island. The area has a subtropical desert climate, with summer temperatures reaching about 42°C and winter averages around 20.1°C. The mean annual temperature is approximately 32.3°C (Al-Shaibani, 2013). Relative humidity varies markedly throughout the year, rising to nearly 70 % in December and falling to about 23 % in June. These seasonal shifts affect evapotranspiration and limit opportunities for groundwater recharge (Benaafi et al., 2025).

Seven main geological units crop out in Al-Qatif, including the Dammam, Hadruk, and Dam formations, together with extensive Quaternary deposits (Rasheeduddin et al., 1989). Among these, the Dammam Formation is the most prominent. It consists mainly of consolidated and dolomitized limestone with marl layers. The Alat Member is characterized by porous dolomitized limestone, while the Al-Khobar Member comprises fractured and karstified limestone and forms the primary water-bearing unit of the Al-Khobar aquifer. The lower part of the Dammam Formation contains shale and marl beds that function as an aquitard and limit vertical groundwater flow. The Hadruk Formation includes alternating shale and sandstone layers, whereas the Dam Formation is made up of marine clays, limestone, and marl (Edgell, 1997).

The Al-Qatif area contains three principal aquifer systems. These include the shallow Neogene aquifers and the deeper Alat and Al-Khobar aquifers. The Alat aquifer forms the upper section of the Dammam Formation. Its thickness averages around 25 m, although it varies locally. Reported transmissivity values range from $2.6 \times 10^{-5} \text{ m}^2/\text{s}$ to $5.1 \times 10^{-3} \text{ m}^2/\text{s}$ (Al-Shaibani, 2013). The Al-Khobar aquifer consists of well-developed karstified limestone with high productivity. It attains a thickness of up to 80 m and supplies good-quality irrigation water to the Al-Qatif and Al-Hassa regions. Its transmissivity ranges between $0.9 \times 10^{-2} \text{ m}^2/\text{s}$ and $0.29 \text{ m}^2/\text{s}$ (Abderrahman et al., 1995).

2.2. Physicochemical data collection and analysis

Groundwater was sampled at thirty-nine locations distributed across the Al Qatif area. Water was pumped from selected wells using submersible pumps. Each well was purged for about fifteen minutes prior to sampling in order to remove stagnant water and obtain a representative sample (Roll et al., 2016). Static water levels in both shallow and deep aquifers were measured with a portable water level meter and referenced to mean sea level. Field measurements of physicochemical properties were carried out on site using a multiparameter instrument to ensure reliable readings at the time of collection. Samples were passed through $0.45 \mu\text{m}$ membrane filters to remove suspended particles and then stored at 4°C before laboratory analysis. For each location, two 500 Millilitre polyethylene bottles were filled for separate cation and anion determinations. Thirteen physicochemical variables were determined at every sampling point. These include pH, salinity (mg/L), sodium (Na^+ , mg/L), potassium (K^+ , mg/L), magnesium (Mg^{2+} , mg/L), calcium (Ca^{2+} , mg/L), fluoride (F^- , mg/L), chloride (Cl^- , mg/L), bromide (Br^- , mg/L), nitrate (NO_3^- , mg/L), sulfate (SO_4^{2-} , mg/L), bicarbonate (HCO_3^- , mg/L), and turbidity (NTU). Salinity represents the total concentration of dissolved salts in groundwater and was used as the target variable.

3. Methodology

3.1. Data preprocessing

Data preprocessing is a significant step for developing robust ML models (Kang and Tian, 2018; Lebawi et al., 2025; Usman et al., 2025b). Prior to model development, all variables were normalized to the range $[0, 1]$ to ensure numerical consistency across predictors. No missing records were found in the data. The data was then split into training and testing phases, with synthetic data generated exclusively from the training data and real observations reserved for model evaluation.

3.2. Synthetic data generation

3.2.1. Gaussian Mixture Models (GMMs)

Synthetic data generation is essential when the quantity of real data is not sufficient to enable ML modeling (Kuchin et al., 2020; Yan et al., 2024). It is required in areas where data are scarce or obtaining them is unaffordable. GMMs are the foundational algorithms for generating synthetic data that replicates the statistical trends and distributions of the dataset. The models employ probabilistic models that represent a dataset as a combination of multiple Gaussian distributions (Reynolds, 2009). These properties make GMMs exceptionally at modeling the complexity of multimodal distributions. It also enables synthetic data to maintain variability and structure of experimental data. The following expressions provide the probability density function of the GMMs and specify the form of each multivariate Gaussian distribution used in the mixture:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \cdot \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k) \quad (1)$$

where $p(\mathbf{x})$ is the overall probability, K represents the Gaussian components, and π_k is the coefficient for the k - th.

$$\mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k) = \frac{1}{\sqrt{(2\pi)^d |\Sigma_k|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_k)^T \Sigma_k^{-1} (\mathbf{x} - \mu_k)\right) \quad (2)$$

where μ_k is the mean vector, Σ_k represents the covariance matrix, and d is the dimensionality of $\mathbf{x}$.

3.2.2. Jensen-Shannon Divergence (JSD)

The JSD is applied to assess the consistency and adequacy of synthetic datasets. It serves as a statistical measure that quantifies the degree of similarity (Sáez et al., 2017). As a symmetrical metric, JSD is set to compare probability distributions for the alignment of

synthetic and real data. The average JSD score was computed to provide an overall measure of similarity for validating the generated synthetic dataset, as shown in the following equation:

$$JSD(P||Q) = \frac{1}{2}KL(P||M) + \frac{1}{2}KL(Q||M) \quad (3)$$

where $M = \frac{1}{2}(P+Q)$ is the average distribution of P and Q and $KL(P||M)$ represents the Kullback-Leibler (KL) divergence.

The hyperparameters of the GMM were defined using explicit selection criteria based on the training data, as summarized in Table 1. The number of mixture components was determined by minimizing the Bayesian Information Criterion (BIC) using the training data only, while all other parameters were fixed to avoid unnecessary model complexity. Post-generation clipping based on training quantiles was applied to ensure that synthetic samples remained within physically realistic bound.

3.2.3. Maximum Mean Discrepancy (MMD)

Maximum Mean Discrepancy (MMD) is used to evaluate whether the synthetic data preserves the multivariate structure of the original groundwater dataset. Unlike JSD, which assesses marginal similarity, MMD measures the distance between joint probability distributions. This makes MMD suitable for detecting differences in dependence structure among hydrochemical variables. Given two datasets, P (real data) and Q (synthetic data), the MMD is defined as:

$$MMD^2(P, Q) = \mathbb{E}_{x, x' \sim P}[k(x, x')] + \mathbb{E}_{y, y' \sim Q}[k(y, y')] - 2 \mathbb{E}_{x \sim P, y \sim Q}[k(x, y)] \quad (4)$$

where $k(\cdot, \cdot)$ is a positive-definite kernel function.

3.2.4. Charge Balance Error (CBE)

Charge Balance Error (CBE) is used to assess the hydrochemical consistency of the synthetic groundwater samples. While statistical metrics evaluate distributional similarity, CBE provides a geochemically grounded check by examining whether the generated data preserves the ionic balance characteristics of the observed groundwater. The CBE is computed as:

$$CBE(\%) = 100 \times \frac{\sum \text{Cations (meq/L)} - \sum \text{Anions (meq/L)}}{\sum \text{Cations (meq/L)} + \sum \text{Anions (meq/L)}} \quad (5)$$

where major ions are measured in milliequivalents per liter. The absolute value of CBE is used to evaluate the degree of ionic imbalance.

CBE statistics were computed for both real and synthetic datasets to compare their behavior. The objective was not to enforce ideal electroneutrality, but to verify that the synthetic samples reproduce the same level of ionic balance and uncertainty present in the observed groundwater data. This ensures that synthetic data generation does not introduce unrealistic chemical artifacts that could bias ML model training.

3.3. Machine Learning (ML) models

3.3.1. Light Gradient Boosting Machine (LightGBM)

LightGBM is fast gradient-boosting framework developed by Microsoft in 2017 (Ke et al., 2017). It has garnered considerable attention since its inception due to its ability to handle large and high-dimensional data. Light as the name suggests, its lightweight structure provides incredible training rates with considerably less memory over typical gradient boosting frameworks. The model also supports categorical features, hence, minimizes memory usage and provides scalability with parallel learning (Cai et al., 2023).

3.3.2. Extreme Gradient Boosting (XGBoost)

XGBoost is a general-purpose ML algorithm with enhanced performance in predictive modeling. It works by expanding on the gradient boosting algorithm applied on the previous one and in preventing overfitting and sparsity for the missing values (Chen and Guestrin, 2016). XGBoost employs an advanced tree-pruning algorithm and a minimum loss reduction requirement to build optimum trees in comparison to traditional gradient boosting algorithms. It also employs scalability through parallel computing as an advanced feature (Velarde et al., 2023).

3.3.3. Random Forest (RF)

Random Forest is an ensemble ML method used for both classification and regression tasks (Svetnik et al., 2003). During training, it

Table 1
Hyperparameter selection and settings used of GMM for synthetic data generation.

Hyperparameter	Candidate / Search space	Selection rule	Final value
Number of components (K)	$K \in \{1, 2, \dots, 8\}$	Minimize BIC on training data	$K = 8$ (lowest BIC)
Covariance type	{full}	Fixed	full
Random state	—	Fixed	42
Clipping of synthetic samples	Train quantiles (1st–99th)	Fixed post-processing	Applied to all variables

constructs a large number of decision trees and aggregates their outputs to produce a more stable and reliable prediction. Its strength and higher performance from two techniques: bootstrap aggregating (bagging) and feature randomness (Lee et al., 2020). The bagging trains enable RF to handle high-dimensional datasets, noisy data, and irrelevant features.

3.3.4. Gradient Boosting Machine (GBM)

GBM is a model that stacks multiple simple models or decision trees together to create a more robust model (Daning et al., 2023; Zhang and Haghani, 2015). In contrast to RF, the GBM model trains trees in a sequential manner whereby decision trees are trained sequentially. It emphasizes data points that are more difficult to predict by accentuating the residual error. This enables the model to capture complicated and non-linear relationships within data (Ilhan et al., 2023). The main benefit of GBM is to provide flexibility to be tailored using hyperparameters and can be tuned for optimal performance (Ayyadevara, 2018).

The hyperparameter configurations used for the ML models are summarized in Table 2. For each algorithm, a consistent number of trees was adopted to allow fair comparison, while learning rate and tree depth were selected to balance model flexibility and overfitting risk. Parameters related to data subsampling, regularization, and tree structure were fixed to commonly accepted values based on prior applications in environmental modeling. All models were trained on the same random seed for uniform performance over repeated runs and for replicable evaluation. Model performance was tested with a separate leakage-free train-test split using fixed random seeds. Cross-validation was not applied due to the limited number of real samples, and stability was instead inferred from consistent generalization behavior across models.

3.3.5. Performance evaluation

Performance evaluation is key to ascertaining the reliability and accuracy of ML models (Jibrin et al., 2024; Uchenna et al., 2025; Usman et al., 2025b). It involves the use of several metrics in quantifying the degree of agreement between predicted and observed values. Model evaluation was performed on a real-only test subset, and synthetic samples were generated exclusively from the training subset to avoid data leakage. The following are the performance metrics used in the study:

- i. Relative Root Mean Squared Error (rRMSE):

$$rRMSE = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (S_i - \hat{S}_i)^2}}{\bar{S}} \quad (6)$$

- ii. Relative Mean Absolute Error (rMAE):

$$rMAE = \frac{\frac{1}{n} \sum_{i=1}^n |S_i - \hat{S}_i|}{\bar{S}} \quad (7)$$

- iii. Weighted Mean Absolute Percentage Error (wMAPE):

$$wMAPE = \frac{\sum_{i=1}^n |S_i - \hat{S}_i|}{\sum_{i=1}^n S_i} \quad (8)$$

- iv. Symmetric Mean Absolute Percentage Error (sMAPE):

$$sMAPE = \frac{1}{n} \sum_{i=1}^n \frac{|S_i - \hat{S}_i|}{\frac{|S_i| + |\hat{S}_i|}{2}} \cdot 10 \quad (9)$$

where S_i is the observed salinity value, $\hat{S}_i$ represents the predicted value, $\bar{S}$ and $\bar{\hat{S}}$ stands for the mean of the observed and predicted values, and n represents the total number of data points.

Table 2

Hyperparameter settings used for ML models in salinity prediction.

Model	Final setting
LightGBM	100 trees; learning rate = 0.10; unlimited depth; minimum leaf samples = 20; random seed = 42
XGBoost	100 trees; depth = 6; learning rate = 0.30; subsampling = 1.0; column sampling = 1.0; L2 regularization = 1.0; random seed = 42
RF	100 trees; minimum split samples = 2; minimum leaf samples = 1; random seed = 42
GBM	100 trees; learning rate = 0.10; depth = 3; random seed = 42

3.4. SHapley Additive exPlanations Analysis (SHAP)

SHAP is an eXplainable Artificial Intelligence (XAI) technique that explains the performance of ML models through assigning the contribution of each feature to predictions. SHAP enables interpretability by assigning the result of predictions to input features to enhance transparency in their significance assessment (Jibrin et al., 2025). The main advantage of SHAP is its ability to provide consistent interpretation of interaction effects in individual predictions. SHAP differs from traditional feature importance metrics by estimating the effect size and direction (positive or negative) of each feature. Fig. 1 presents the research methodology flowchart used in this study.

$$\hat{y} = \phi_0 + \sum_{i=1}^M \phi_i \quad (10)$$

where ϕ_0 is the model's baseline (expected) prediction and ϕ_i is the SHAP value for feature i .

SHAP interaction values $\phi_{i,j}$ quantify the contribution of the interaction:

$$\phi_{i,j} = \text{SHAP value of feature interaction between } i \text{ and } j \quad (11)$$

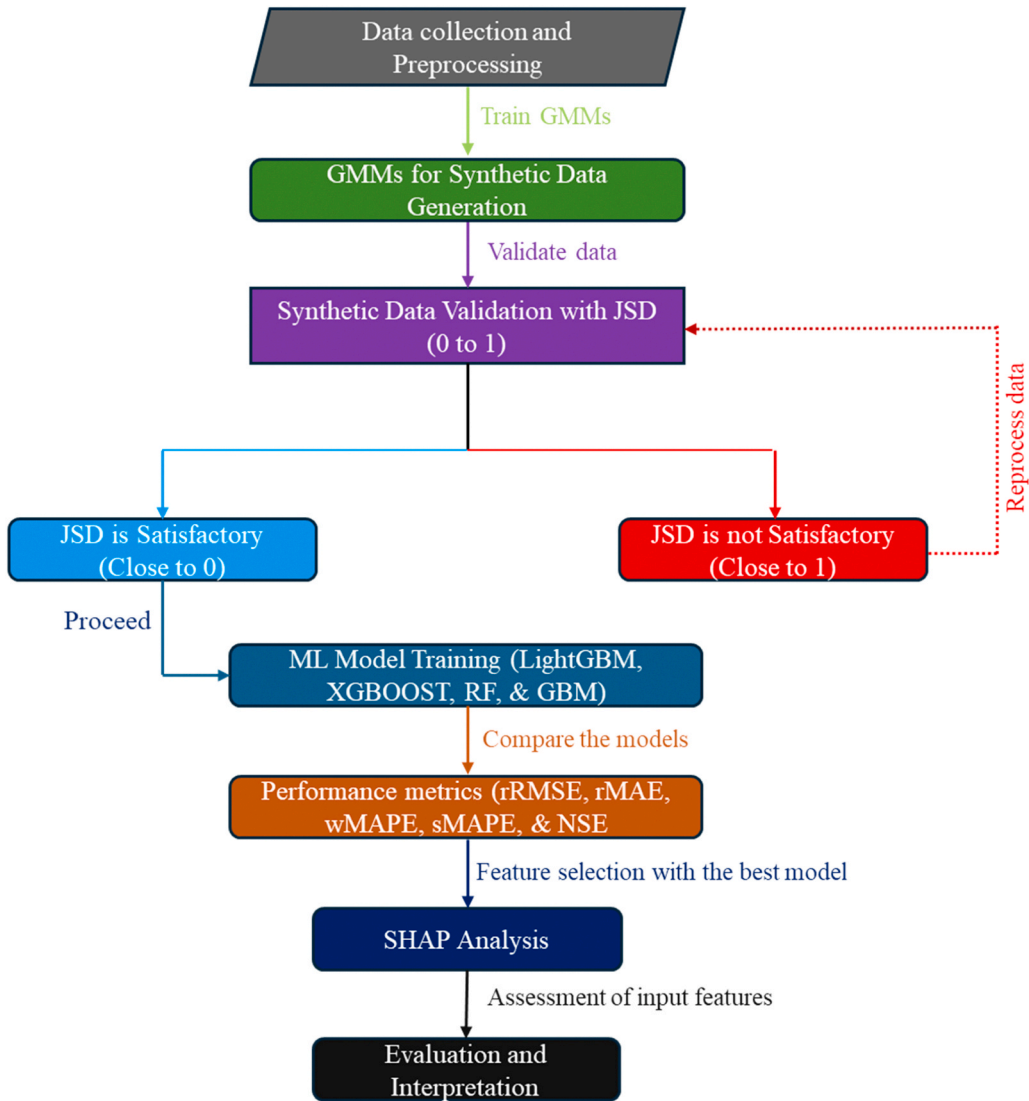


Fig. 1. Workflow of the proposed framework for groundwater salinity prediction, including synthetic data generation, validation, model training, and interpretability analysis.

4. Results and discussion

4.1. Dataset partitioning and synthetic data augmentation strategy

The groundwater dataset was first divided into training and testing subsets to ensure an unbiased assessment of model performance. Approximately 70 % of the real observations were assigned to training, while the remaining 30 % were reserved for testing and were not involved in any stage of synthetic data generation. Synthetic samples were produced using GMMs fitted exclusively to the real training data and were combined only with the training subset. All predictive evaluations were carried out on real test observations, thereby avoiding information leakage. The number of Gaussian components used in the GMM was selected using the BIC. The BIC provides a quantitative balance between goodness of fit and model complexity by penalizing excessive numbers of parameters (Fatollahi and Alishahi, 2025; Luo, 2023). The lower BIC values indicate more suitable representation of the data structure. In Fig. 2, the BIC value declined sharply as the number of components increased from one to three, dropping to about 2030 at three components from approximately 3100 at one component. This extensive reduction means that a single or two-component model could not be used to capture the multimodal structure of hydrochemical data. The reduction was even greater when the model was reduced to 4 or 5 components, with the BIC reaching approximately 1270 at 5 components. After that, the improvement became less consistent. The BIC increased slightly to six components and rose more noticeably to seven components. The lowest BIC value, approximately 1120, was obtained for eight components, indicating the best compromise between model fit and parsimony within the tested range. Based on this objective criterion, a GMM with eight components was selected for synthetic data generation. Using this configuration, 700 synthetic samples were generated from the training data distribution, resulting in an augmented training dataset of 727 samples.

Fig. 3 shows that the GMM-generated synthetic data closely reproduce both the mean values and the variability of the real training dataset across most hydrochemical variables. The alignment of mean concentrations indicates that the synthetic samples preserve the central tendency of the observed groundwater chemistry, while the comparable standard deviations demonstrate that the spread and heterogeneity of the data are largely maintained. The strongest agreement is observed for major ions such as sodium, calcium, magnesium, and sulfate. Slight smoothing is evident for high-variance variables, particularly chloride and salinity, indicating the probabilistic nature of the mixture model rather than distortion of the underlying hydrochemical structure.

4.2. Validation of synthetic data quality

4.2.1. Marginal distribution similarity

The JSD was employed to compare the marginal distributions of the original training dataset with those of the generated synthetic

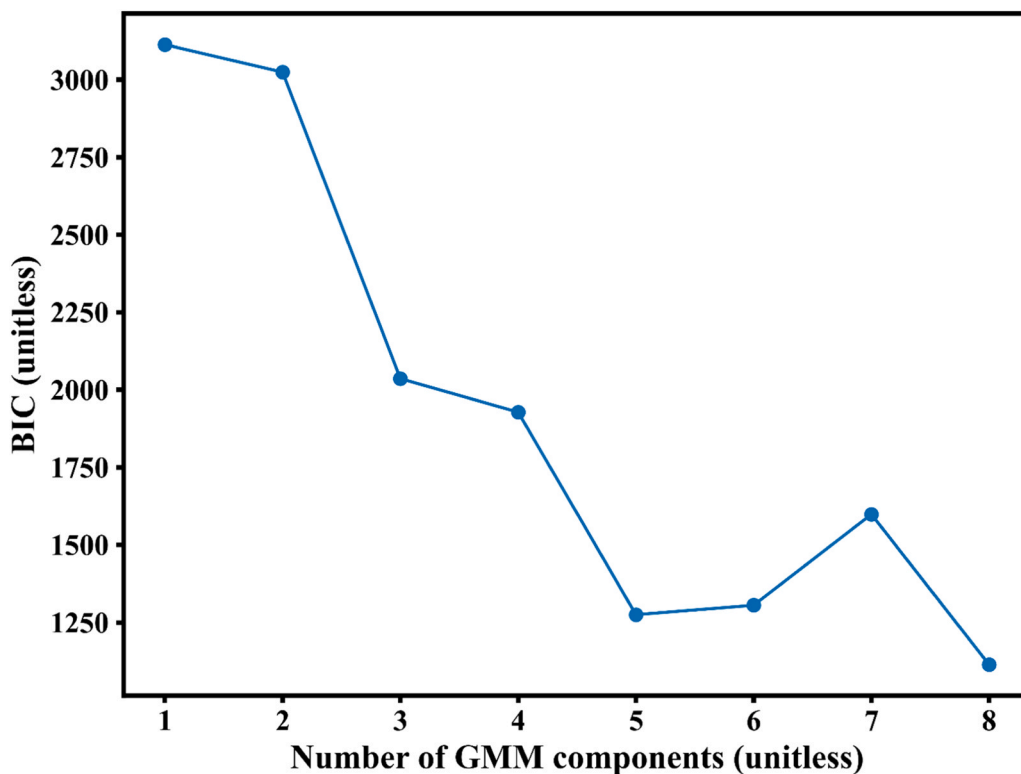


Fig. 2. Selection of the optimal number of Gaussian mixture components based on the BIC.

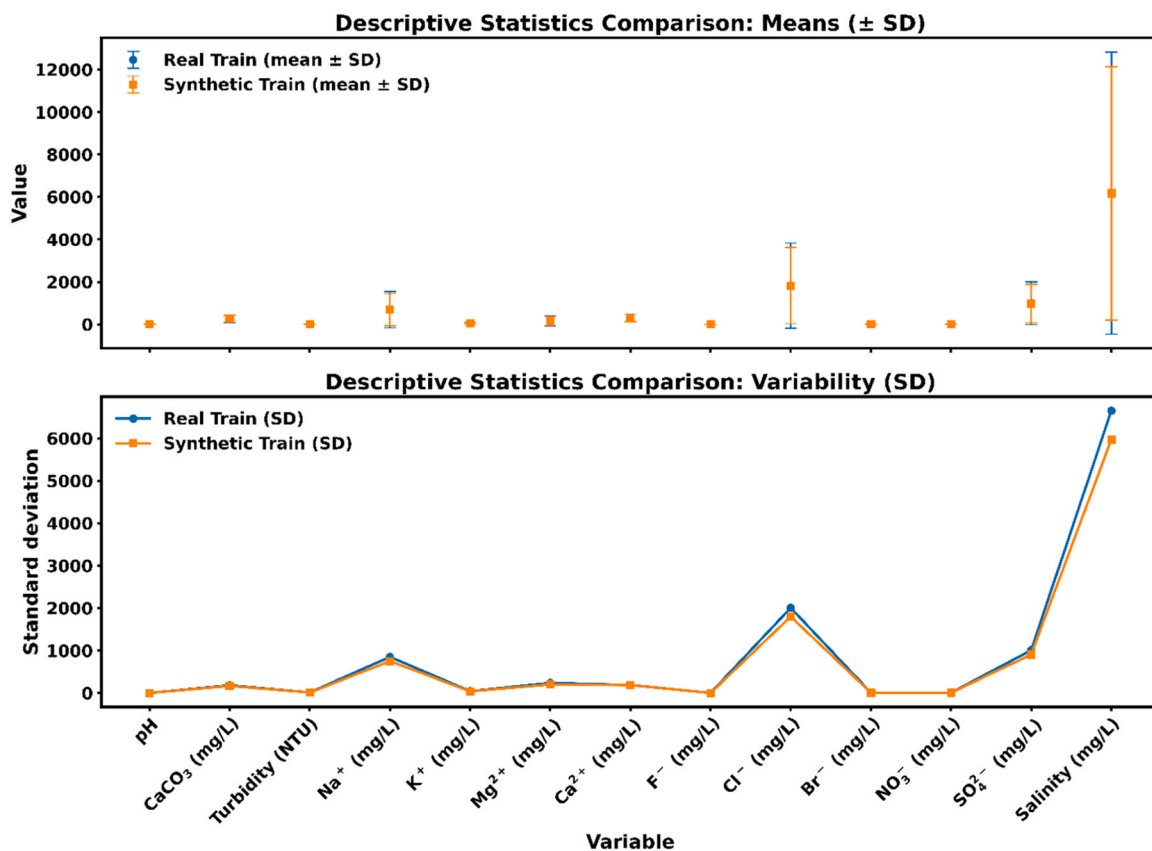


Fig. 3. Comparison of descriptive statistics between real and synthetic training datasets.

samples. It quantifies the distance between probability distributions and is a bounded, symmetric measure, with values equal to zero indicating identical distributions and lower values corresponding to greater similarity. The average JSD across all the variables in this dataset was 0.0797, indicating a smaller overall difference between observed data and synthetic data. As shown in Fig. 4, the JSD for most variables is well below 0.10, indicating that the synthetic samples accurately imitate the univariate features of the real training data. Chloride and magnesium showed the least deviation, both have JSD values close to 0.03. Although these variables exhibit relatively stationary distributions, they are largely characterized by the Gaussian mixture structures for the models. Low JSD values for

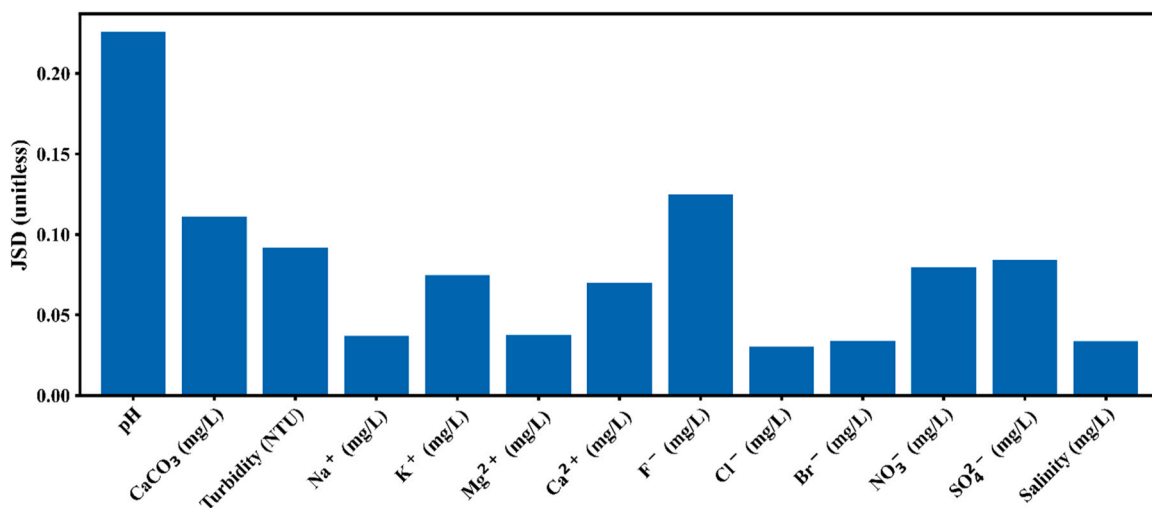


Fig. 4. Marginal distribution similarity between real training data and synthetic samples quantified by the JSD.

sodium and salinity also reflect good representation of concentration ranges important to the salinization process. Values in JSD were higher for pH and fluoride where pH shows its biggest divergence on the range around 0.22. This is representative of the narrower range and more sensitive pH measurements and deviations around small changes of distribution can lead to relatively large divergence values. Fluoride also exhibits a higher JSD, due to its skewed distribution and limited number of distinct concentration levels in the original dataset.

4.2.2. Multivariate structure preservation

While marginal similarity assesses individual variables independently, preservation of multivariate structure is essential for hydrochemical datasets where strong dependencies exist among ions. To evaluate whether these joint relationships were retained, the similarity between the real training data and the synthetic samples was assessed using the MMD with a radial basis function kernel. The computed MMD value was 0.0, indicating no detectable difference between the joint distributions of the real and synthetic training datasets under the selected kernel. This result suggests that synthetic samples preserve the dependence structure among hydrochemical variables. The MMD measures differences between probability distributions as a symmetric, non-negative measure that equals zero when the distributions are identical (Gretton et al., 2012). The kernel bandwidth parameter, $\gamma = 0.0589$, was determined using the median heuristic and provides sensitivity to both local and global distributional differences. The zero value of the MMD value does not imply exact duplication of observations but rather statistical indistinguishability between the two datasets in the multivariate space. This outcome reflects the ability of the GMM to represent the joint distribution learned from the training data without introducing artificial correlations or distorting existing relationships. Given the limited size of the original dataset, the results demonstrate that synthetic augmentation did not compromise the internal structure of the hydrochemical system.

4.2.3. Hydrochemical consistency assessment

Apart from statistical correspondence, hydrochemical consistency was evaluated to examine whether the synthetic samples preserved the ionic balance characteristics of the observed groundwater data. CBE was used as a diagnostic measure. It rates the agreement of all the summed major cations and anions in a water sample and low errors indicate the internal chemical behaviour (Atkinson, 2024). Table 3 provides the CBE statistics for real training data and the synthetic training samples. For the real training dataset, the median absolute CBE was 14.39 %, with a mean value of 13.92 %. Only 7.41 % of the samples fell within ± 10 %, and none satisfied the stricter ± 5 % threshold. The observed imbalance reflects the characteristics of the original measurements rather than artifacts introduced during data processing.

The synthetic training data exhibited comparable charge balance behavior. The median absolute CBE was 14.06 %, and the mean value was 13.85 %, both slightly lower than those of the real training data. The proportion of samples within ± 10 % increased marginally to 9.29 %, while 0.86 % of the samples fell within ± 5 %. These differences are minor and do not indicate a systematic improvement or deterioration of ionic balance. Instead, they suggest that the synthetic data reproduced the inherent hydrochemical characteristics of the real dataset, including its limitations. Importantly, the synthetic samples did not exhibit exaggerated charge imbalance or artificially enforced electroneutrality. Preserving the original level of imbalance is preferable in this context, as forcing ideal charge balance would introduce unrealistic chemical behavior and distorting ion relationships learned from the observed data. The strong agreement between the real and the synthetic CBE statistics indicates the GMM can preserve hydrochemical stability while creating more samples. In general, the CBE analysis confirms that the synthetic data reflect the same hydrochemical constraints present in the real training data.

4.2.4. Hydrochemical signatures of seawater intrusion

Classical hydrochemical indicators were examined to provide process-based evidence of seawater intrusion. The median Na^+/Cl^- ratio of 0.56 (meq/meq) indicates a dominant chloride contribution relative to sodium, consistent with conservative mixing with saline water rather than evaporite dissolution (Klassen et al., 2014; Zarei et al., 2013). A high median Cl^-/Br^- ratio of 536 and an exceptionally strong Cl–Br correlation ($r = 0.995$) indicate near-linear co-variation of these conservative ions, which is characteristic of freshwater–seawater mixing in coastal aquifers. The $\text{SO}_4^{2-}/\text{Cl}^-$ ratio of 0.43 and the $(\text{Ca}^{2+} + \text{Mg}^{2+})/(\text{Na}^+ + \text{K}^+)$ ratio of 1.08 further suggest limited sulfate enrichment and partial buffering by alkaline earth cations, consistent with ion exchange and carbonate interactions superimposed on saline intrusion. Concurrently, these ionic ratios provide independent hydrochemical evidence of seawater intrusion as the dominant salinization process in the Al-Qatif aquifer.

4.3. Machine learning model performance

4.3.1. Training performance using augmented data

The predictive performance of the ML models was first evaluated during the training phase using the augmented dataset, which combined real and synthetic samples. This assessment provides insight into how effectively each model learns the underlying

Table 3
Comparison of the CBE statistics between real and synthetic training data.

Dataset	Median CBE (%)	Mean CBE (%)	Samples within ± 5 % (%)	Samples within ± 10 % (%)
Real Train	14.3912	13.9184	0.0000	7.4074
Synthetic Train	14.0596	13.8464	0.8571	9.2857

relationships present in the expanded feature space. All models achieved low error values during training, indicating effective fitting to the augmented data as shown in Table 4. XGBoost exhibited the lowest training errors across all metrics, with an rRMSE of 0.0486 % and an rMAE of 0.0345 %, followed closely by the RF model, which recorded an rRMSE of 1.3832 %. However, these results demonstrate the strong ability of ensemble-based models to address nonlinear relationships when there are enough training samples.

LightGBM also showed stable training efficiency, showing an rRMSE of 2.6028 %, and the GBM, meanwhile, had comparatively higher training errors, with an rRMSE of 0.9783 % and sMAPE of 1.3660 %. The differences among models can be attributed to variations in tree construction strategies, learning rates, and sensitivity to noise introduced through data augmentation. The augmented dataset includes synthetic samples created within the training distribution, which can favor models with high flexibility. Thus, training outputs primarily reflect the model's capacity to learn from the enriched data representation, not its predictive robustness. As a result, model suitability is interpreted based on the testing phase, where performance is evaluated exclusively on real, unseen observations.

4.3.2. Testing performance on real-only observations

Model performance was further evaluated using the test dataset composed exclusively of real groundwater observations. This evaluation provides a direct measure of model generalization because no synthetic samples were used at this stage. As shown in Table 5, model performance varied noticeably when tested on unseen real data. The GBM achieved the lowest prediction error rate (rRMSE = 7.87 %, rMAE = 6.73 %), suggesting the best generalization ability among the models. When all error criteria are evenly distributed, GBM's relatively moderate performance indicates that it reasonably sums all salinity-hydrochemical reaction. GBM is effective in capturing complex, non-linear patterns and generalizing well to unseen data due to their iterative learning approach (Adler and Painsky, 2022). Learning-rate adjustment and stochastic perturbations are also employed in GBM to avoid overfitting (Konstantinov and Utkin, 2021). In a comparative study by (Hindarto, 2024), GBM was slightly more accurate (82 %) compared to LightGBM (81 %), which means that GBM yield better predictive performance.

The RF model also had competitive performance with rRMSE 10.02 % and rMAE 7.42 %. Although these errors are higher than those of GBM, the results indicate reasonable robustness to real observations. LightGBM and XGBoost had more errors as XGBoost obtained the maximum rRMSE (12.21 %) and sMAPE (10.64 %) among the models. The discrepancy between training and testing performance highlights the importance of leakage-free validation. While some models benefited strongly from the augmented training dataset, their predictive capability diminished when confronted with real-only observations. This behavior reflects differences in model sensitivity to data augmentation and their ability to handle limited, noisy hydrochemical measurements. In general, the testing results confirm that GBM provides the most reliable balance between learning capacity and generalization. Beyond predictive accuracy, model behavior shows robustness to data perturbation. RF and GBM showed stable convergence with lower computational demand. The sensitivity of LightGBM and XGBoost is reflected in their larger gaps between training and real-only testing performance. On the other hand, GBM maintained a more balanced error profile.

Fig. 5 provides a comparative view of model performance based on the four evaluation criteria using a radar plot. The values were normalized to enable comparison of rRMSE, rMAE, wMAPE, and sMAPE, where larger radial extents correspond to larger error magnitudes. It enables clear comparisons of error profiles on the real-only test data of the models. The LightGBM model shows the largest enclosed area on all four axes. XGBoost shows a similarly expanded profile, particularly along the rRMSE and sMAPE axes, confirming its sensitivity to unseen observations. On the other hand, the RF model displays a more compact and balanced shape, with moderate error values across all metrics. Its radar footprint is noticeably smaller than those of LightGBM and XGBoost, suggesting improved robustness under real-data testing conditions. The GBM demonstrates the most compact polygon among all models, with reduced radial values across all four metrics. This pattern indicates the lowest overall error burden and a more stable trade-off among different error measures. The radar plot (See Fig. 5) reinforces the conclusions drawn from the numerical evaluation by providing an integrated view of model behavior across multiple performance dimensions.

4.4. Explainable AI analysis of salinity drivers

4.4.1. Global feature importance and contribution patterns

The global contribution of input variables to groundwater salinity prediction was examined using SHAP. Fig. 6 presents the SHAP summary beeswarm plot. The plot illustrates both the relative importance of each feature and the direction of its influence on model predictions (Aldrees et al., 2025a; Yayavaram et al., 2024). Each point represents an individual test observation, positioned according to its SHAP value, while the color scale indicates the magnitude of the corresponding feature value. The beeswarm plot reveals that Br⁻ and sodium Na⁺ exert the strongest influence on salinity predictions. Positive SHAP values for the two variables are higher in both concentrations, which suggest higher salinity under the analysis from the observed data. Its direct salinity contribution to salinity is

Table 4
Training performance of the ML models using augmented data.

Model	rRMSE (%)	rMAE (%)	wMAPE (%)	sMAPE (%)
LightGBM	2.6029	0.4804	0.4804	0.5445
XGBoost	0.0486	0.0344	0.0345	0.0590
RF	1.3832	0.3937	0.3937	0.6272
GBM	0.9783	0.7008	0.7008	1.3660

Table 5
Generalization performance of ML models using real test data.

Model	rRMSE (%)	rMAE (%)	wMAPE (%)	sMAPE (%)
LightGBM	8.2669	6.2513	6.2513	7.7462
XGBoost	12.213	9.6070	9.6070	10.6419
RF	10.0207	7.4157	7.4157	9.2417
GBM	7.8720	6.7318	6.7318	8.4362

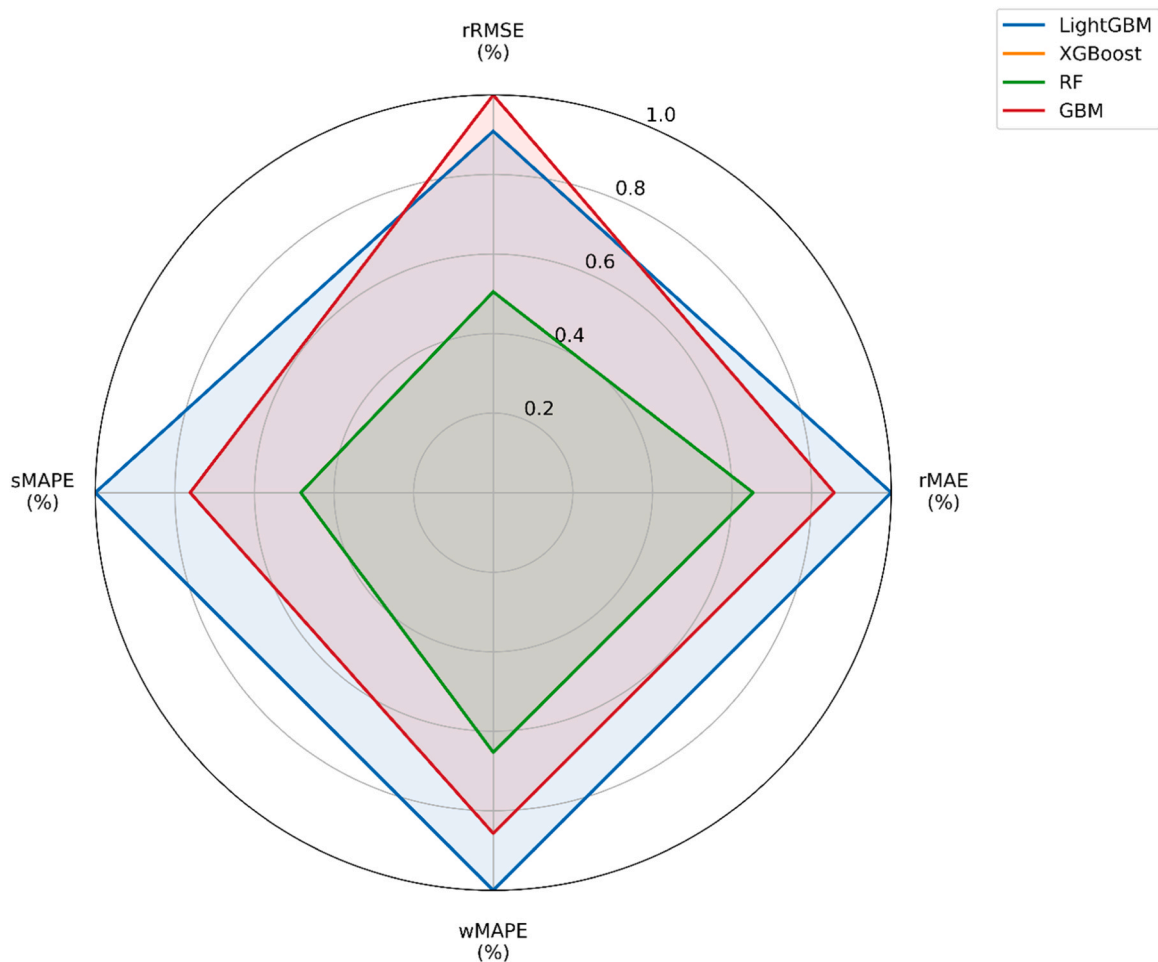


Fig. 5. Radar-based multi-metric comparison of ML model performance on real-only test data.

also significantly higher for Cl^- but less than that of bromide and sodium, though Cl^- is observed to exert a strong positive role in salinity. Large chloride concentrations enhance the predicted salinity and are negative with respect to predictability for chloride, and lower chloride concentrations negatively. Na^+ , Cl^- , and Br^- exhibit coordinated behaviour that is consistent with shared origins and conserving characteristics of transport in saline groundwater systems. Systematic variation of bromide concentrations and Cl^-/Br^- ratios with salinity indicates a combined salt content in freshwater, seawater, and agriculturally-driven waters (Alessandrino et al., 2023).

Ca^{2+} and Mg^{2+} have moderate contributions and positive and negative SHAP values depend on concentration. This dual influence implies that, by different mechanisms, these ions affect the salinity, such as dissolution of the carbonate and exchange of cation. The influence of magnesium and sodium on salinity supports existing research linking these ions to groundwater systems characterized by elevated salinity levels (Hakami et al., 2024; Rossi et al., 2023). CaCO_3 and SO_4^{2-} play a less important role in the model predictions. Salinity is also affected indirectly by CaCO_3 by buffering effects and mineral equilibrium, while sulfate has a low directional effect. F^- , pH, turbidity, K^+ , and NO_3^- have SHAP values clustered close to zero and thereby make a limited contribution to the salinity prediction in the study area.

Table 6 presents the mean absolute SHAP values for the variables and provides information on the quantitative significance.

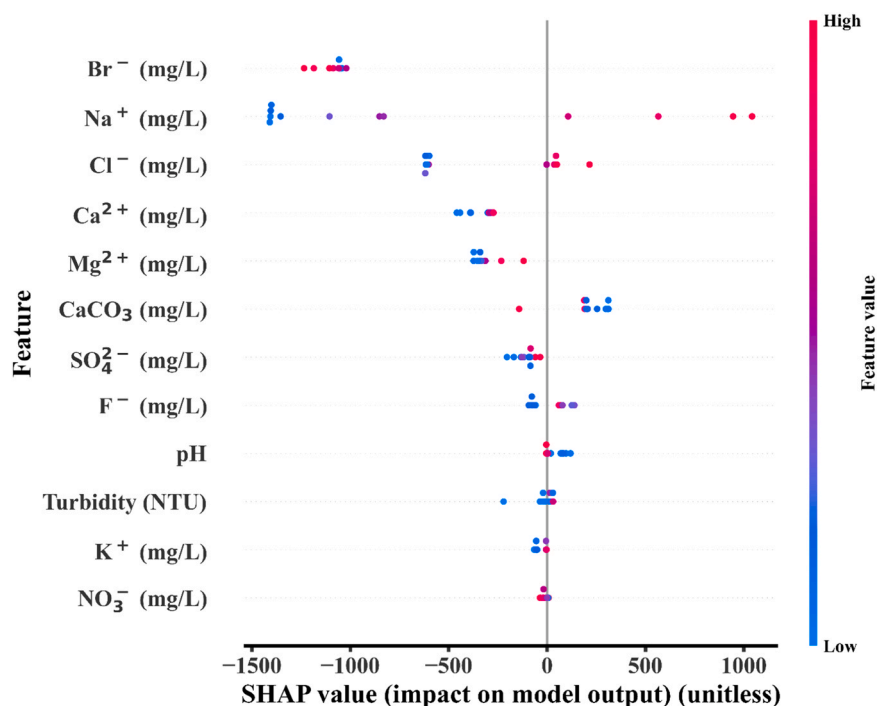


Fig. 6. SHAP Beeswarm plot showing global contribution patterns of hydrochemical variables to salinity prediction.

Bromide ranks first with a mean $|\text{SHAP}|$ value of 1083.66, followed closely by sodium (1035.36). Chloride occupies the third position (385.36), while calcium (331.26) and magnesium (314.16) form a secondary group of influential predictors. Elevated contributions of bromide, sodium, and chloride reflect seawater intrusion in the Al-Qatif aquifer, where over-extraction reduces hydraulic gradients and promotes mixing between saline and freshwater endmembers. A similar dominance of these ions has been reported in Al-Qatif groundwater, where high transmissivity and coastal proximity intensify salinization under sustained pumping (Jibrin et al., 2025). Calcium carbonate (235.01) and sulfate (106.21) suggest relatively lower but still detectable levels of salt mineralization. Calcium carbonate has an effect on salinity by buffering pH, affecting water hardness, and contributing to the ionic balance (Isa et al., 2012). The remaining variables show significantly smaller mean $|\text{SHAP}|$ values, confirming their limited impact on the model output.

4.4.2. Local and global contribution patterns of input features

The SHAP heatmap was used to visualize local and global contribution patterns simultaneously, examining how input variables contribute to salinity predictions across individual observations. Fig. 7 presents the SHAP values of the most influential features for the test samples, where rows represent individual test observations sorted by total absolute SHAP magnitude and columns represent hydrochemical variables. Color intensity is indicative of nature and magnitude of contribution of features in both directions, with positive values representing increasing predicted salinity and negative values pointing toward reduced impact. Br^- and Na^+ dominate most test samples in heatmap. These variables present significant positive and negative contributions in relation to the concentration of

Table 6
Mean absolute SHAP values of hydrochemical variables for salinity prediction.

Feature	Mean $ \text{SHAP} $ (Unitless)
Br^-	1083.658
Na^+	1035.359
Cl^-	385.3612
Ca^{2+}	331.2628
Mg^{+2}	314.1581
CaCO_3	235.0094
SO_4^{2-}	106.2083
F^-	82.6643
pH	53.18584
Turbidity	35.90634
K^+	33.79281
NO_3^-	14.10661

each respective variable. Cl^- and CaCO_3 present moderate but consistent contributions from multiple samples. Their influence is less pronounced than that of Br^- and Na^+ , yet still systematic, implying a secondary control associated both with the water mixing as well with carbonate equilibrium. Ca^{2+} and Mg^{2+} have more localized effects, and their contributions range in significance from positive to negative values of varying values. Such variation mirrors the contributions of these ions to several geochemical processes, including mineral dissolution and ion exchange. On the other hand, SO_4^{2-} , F^- , pH, turbidity, K^+ , and NO_3^- make relatively small contributions in nearly all samples. Their SHAP values cluster around zero with limited variation, indicating little influence on salinity prediction.

4.4.3. Local prediction pathways and feature contribution sequences

Fig. 8 illustrates the SHAP decision plot for the most influential test instances. It provides a step-by-step view of how individual hydrochemical variables contribute to the final salinity prediction. Each polyline represents one test sample, identified by its rank and corresponding index (idx) in the test dataset. The index is only a traceable reference to the location of an observation in the test data and to specific prediction paths that can be interpreted. All prediction paths are derived from the same baseline value, based on which the model outputs a value of 6172. Each feature moves up or down the prediction hierarchy based on its SHAP contribution from this baseline. The horizontal axis shows the model output as features are added one by one, in order of their global importance. Na^+ and Br^- take centre stage in the early stages of prediction paths in all instances shown. For the lower-ranked instances (rank 1, idx 8), these features have significant negative effects, reducing predicted salinity to 2000–3000. By contrast, larger ordered instances (rank 8, idx 0) exhibit large positive changes related to high Na^+ and Br^- concentrations, moving predicted salinity toward 5500–6000. This divergence explains most of the separation between low- and high-salinity predictions. Cl^- consistently contributes after Na^+ and Br^- , reinforcing the prediction direction established by these dominant ions. Its effect is moderate but consistent, indicating conservative behavior in saline groundwater. Ca^{2+} and Mg^{2+} introduce additional variability in the mid-section of the paths. Their contributions differ among instances, with some samples showing partial compensation effects that either dampen or amplify the influence of the dominant ions. These variations are responsible for differences among instances with similar Na^+ and Br^- levels.

CaCO_3 and SO_4^{2-} exert smaller but visible adjustments to the predicted salinity (See Fig. 8). Their effects are more localized and do not alter the overall ranking of the instances, but they contribute to fine-scale separation between prediction paths, particularly in the 4500–6000 salinity range. F^- , pH, K^+ , turbidity, and NO_3^- produce only minor shifts near the end of the decision paths, typically on the order of a few tens of units. The ordering of the instances by rank corresponds to their final predicted salinity values. Lower ranks (idx 8, 10, and 3) converge toward lower outputs below 3000, while higher ranks (idx 7 and 0) converge near or above 6000. This systematic separation also shows that salinity regimes are distinguishable primarily by a subset of dominant hydrochemical variables in the model's distribution. The secondary variables provide a deeper consideration of a few predictions per variable. In general, the decision plot indicates that groundwater salinity predictions arise from consistent feature-contribution sequences across samples, rather than from isolated or unstable effects. The explicit labeling of instance indices allows direct linkage between model explanations and individual observations, strengthening confidence in both the transparency and physical plausibility of the prediction process. Table 7 summarizes recent applications of synthetic data generation across different domains. It describes the modeling strategies and methods previously applied. This comparison situates the present research within the literature and clarifies how GMMs differ from

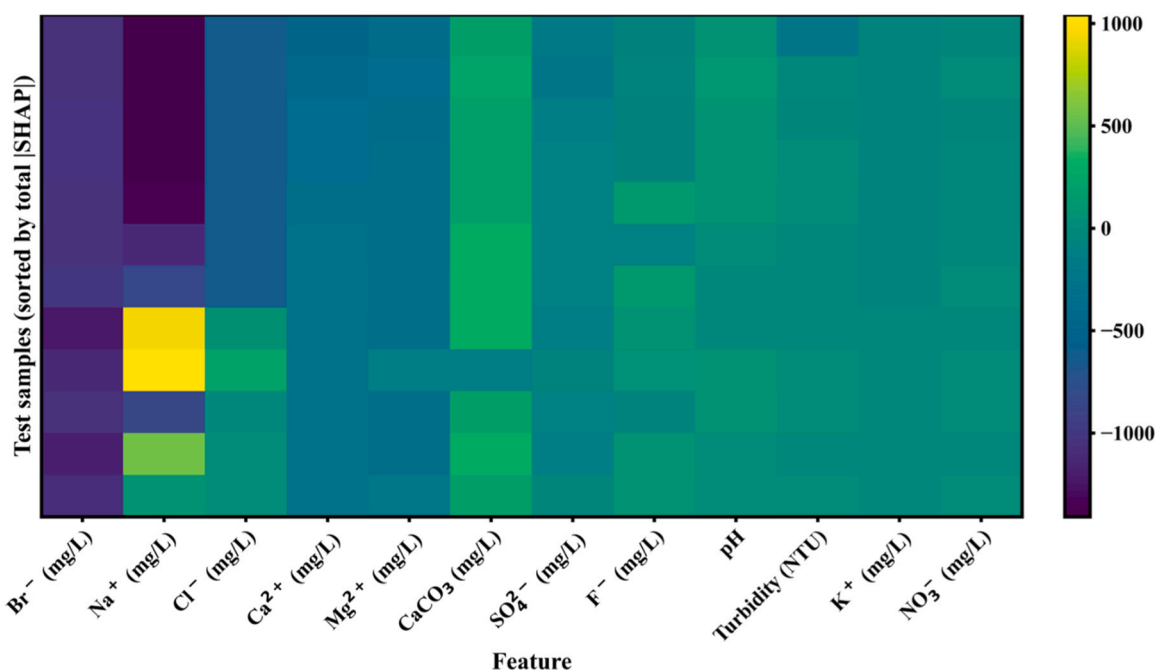


Fig. 7. Local and global SHAP contribution patterns across test samples.

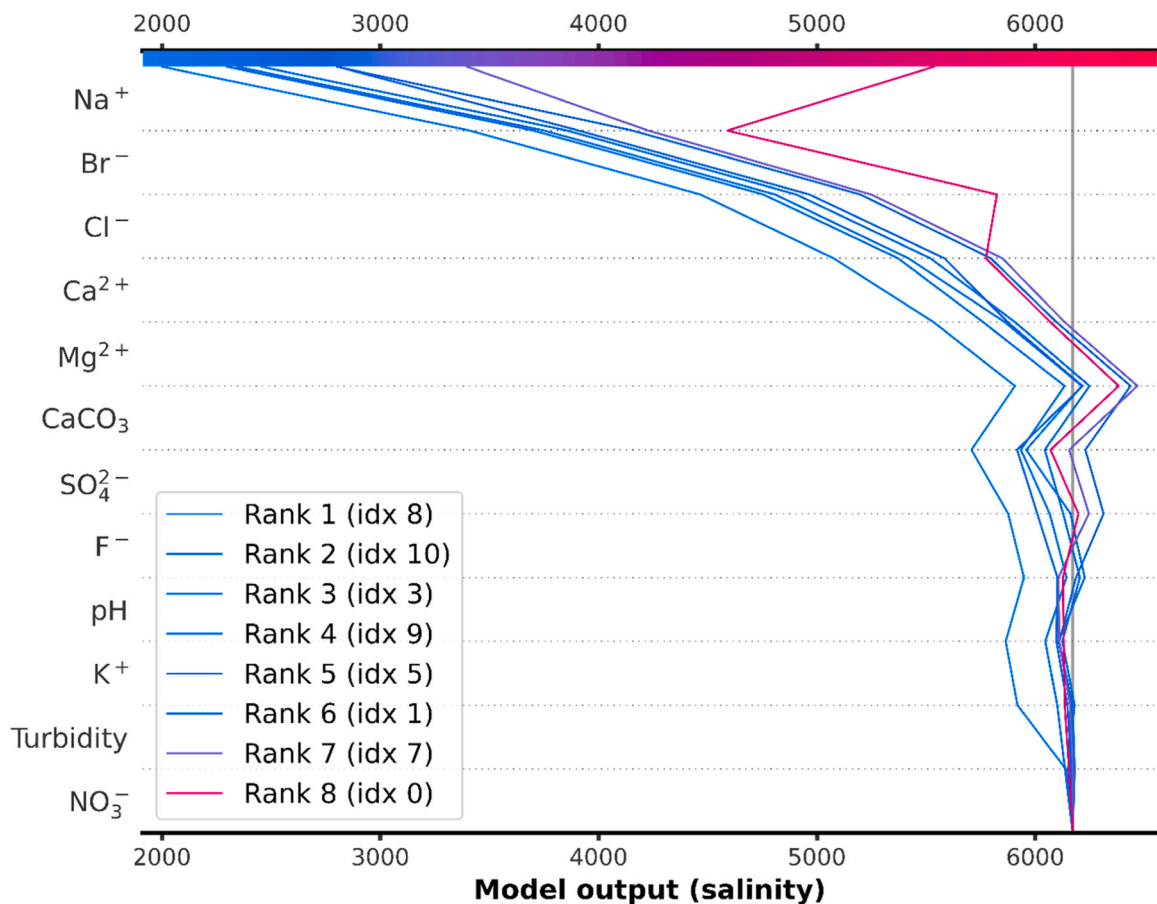


Fig. 8. Decision plot illustrating cumulative SHAP contributions across test instances.

common deep generative and reservoir-based techniques.

5. Conclusions

Groundwater salinity management is increasingly challenged by limited monitoring data and growing pressures from seawater intrusion and over-abstraction. These constraints reduce the reliability of conventional modeling approaches and complicate the application of ML methods that require sufficient. The findings of the current study show that GMMs can be applied to small groundwater salinity datasets to retain some important statistical and hydrochemical features. The synthetic data showed low marginal divergence from observed samples (average JSD = 0.0797), no detectable difference in joint multivariate structure (MMD = 0.0), and CBE statistics comparable to those of the real dataset. These outcomes indicate that the synthetic samples maintain covariance patterns and ionic balance behavior rather than introducing artificial structure. When trained on the augmented dataset and evaluated on real-only test samples, the GBM achieved the most consistent generalization performance. Explainable analysis identified bromide, sodium, and chloride as the dominant controls on groundwater salinity. This reveals seawater mixing and conservative transport processes in the Al-Qatif coastal aquifer.

Some methodological limitations are inherent to the available dataset and are accounted for when interpreting the results. The original dataset comprised only thirty-nine samples, which restricts the representativeness of extreme salinity conditions. Synthetic data generation may therefore smooth out rare high-salinity values. In addition, the findings are specific to the hydrogeological and geochemical conditions of the Al-Qatif aquifer, and extrapolation to other regions requires careful consideration. The proposed framework provides a controlled and interpretable approach to groundwater salinity modeling in data-scarce settings. Its application is most appropriate as a complementary tool to support local groundwater assessment where observational data remain limited.

CRediT authorship contribution statement

Ali Aldrees: Writing – review & editing, Validation, Software, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Salisu Dan’azumi:** Writing – review & editing, Validation, Supervision, Resources, Formal analysis.

Table 7

Comparison of synthetic data generation methods for various applications.

Authors	Method	Study objective	Remarks
(Moeini et al., 2024)	GANs	To model complex dynamics in the hyporheic zone using conditional deep convolutional GANs (cDC-GANs).	The significant of the synthetic data is to explore diverse geological conditions.
(Yang et al., 2024)	GMMs, VAEs, GANs	To enhance social network advertising predictions using synthetic data augmentation techniques.	Compared the effectiveness of generative models in improving classification accuracy. Generated synthetic data based on actual social network datasets.
(Khan et al., 2023)	VAEs	To improve fault classification and localization in power transmission networks by generating synthetic data that captures the statistical properties of real data.	The synthetic data is generated based on real data from power transmission networks to enhance classification accuracy.
(Liu et al., 2023)	ESNs	To develop a novel model (WER-EKSN) for multi-step water level prediction by addressing unstable predictions, parameter dependency, and error accumulation issues.	The synthetic data is generated using ESNs to improve forecasting for water level prediction.
(Yilmaz and Korn, 2022)	GANs	To generate synthetic electricity consumption data for individual users using GAN architectures.	Generated synthetic data by learning from observed electricity consumption data to address privacy concerns, enhance prediction accuracy, and manage grid loads efficiently.
(Kuchin et al., 2020)	GANs	To assess the upper performance limits of ML classifiers in borehole data classification	The synthetic data is centered on observed lithological parameters to evaluate the effectiveness of ML classifiers.
This study	GMMs	To generate synthetic groundwater salinity data from a small observational dataset using GMMs and to support salinity prediction through ML models under data-scarce conditions.	Synthetic data are generated by explicitly modeling the joint statistical structure of hydrochemical variables and are validated using JSD, MMD, and CBE to ensure statistical and hydrochemical consistency prior to model training.

Abdulhayat M. Jibrin: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Mohammad Al-Suwaiyan:** Writing – review & editing, Validation, Supervision, Project administration, Methodology. **Zaher Mundher Yaseen:** Validation, Supervision, Software, Resources, Project administration. **Sani I. Abba:** Writing – review & editing, Project administration, Methodology, Investigation, Data curation.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

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